AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 24 · STUDENT LESSON

Train, Validation, Test

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 8: split by the independent unit before fitting transforms.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Select a model using a declared validation rule. Try the recall before opening notes.

Your learning target

I can select a model using a declared **validation** rule.

I can keep **training**, **validation** and test roles separate.

I can recognize repeated **validation** as development.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Training fits parameters, **validation** chooses settings, and testing estimates final performance after choices are frozen.



Your target

LEARNING TARGET You have 30 leaves with 3 views each. Allocate 18 leaves to train,6 to **validation**,6 to test. How many images are in each? Where is a learned normalisation mean fitted?

Builder name and experiment date:

Readiness for the connected explanation

Which data fitted the line last week?

This week's end goal

By the end, I can explain train, **validation**, test using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Train, Validation, Test case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use train, **validation**, test to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Fitting and selecting answer different questions. **Training** adjusts parameters. **Validation** helps choose settings or model families. A final test estimates the performance of the frozen development **process** on held-out examples.

Keep related records in the same split when separate rows could reveal the same source.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **validation** and model selection.

STUDENT LESSON · 3 OF 18

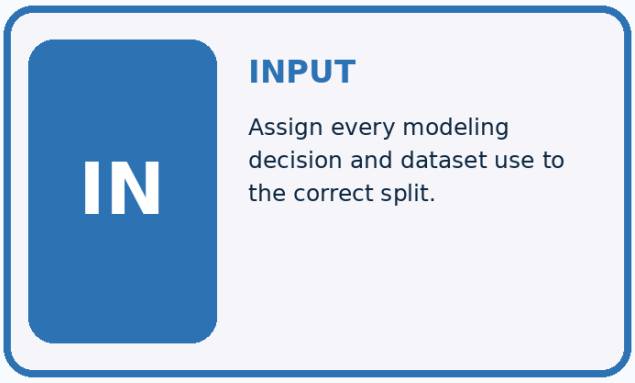
See how it works

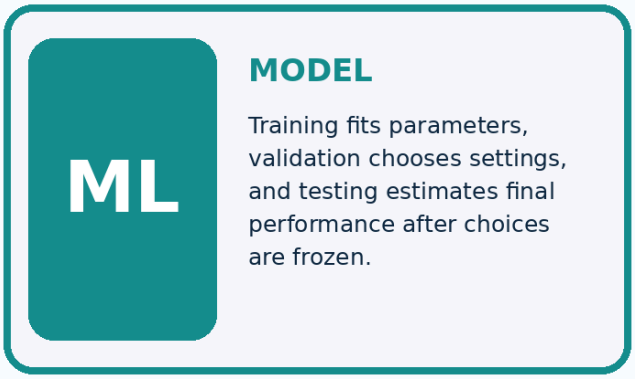
> Follow the mechanism for Train, Validation, Test. Connect each labelled input, deciding step and result.

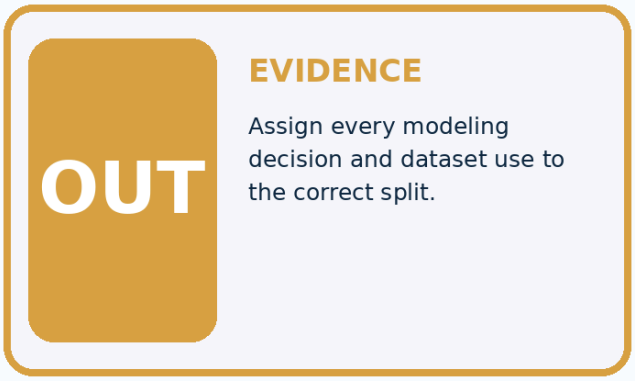
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Assign every modeling decision and dataset use to the correct split.

MODEL IDEA **Training** fits parameters, **validation** chooses settings, and testing estimates final performance after choices are frozen.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Fit two candidate trees on **training** rows. Choose depth and threshold using **validation** metrics and error slices.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Three candidates have **training** MAE A=1, B=0, C=2 and **validation** MAE A=3, B=5, C=3. The declared rule is smallest **validation** MAE, then simpler model, with A simpler than C. Choose A.

B fits **training** perfectly but does worse on **validation**. The final test remains unopened.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

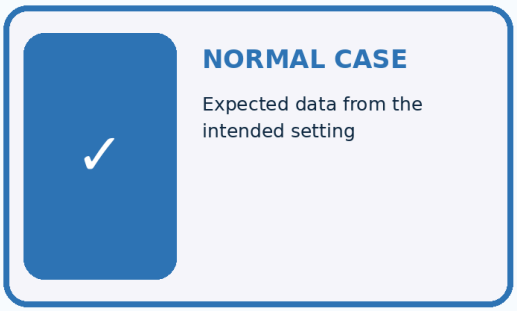
Compare the cases

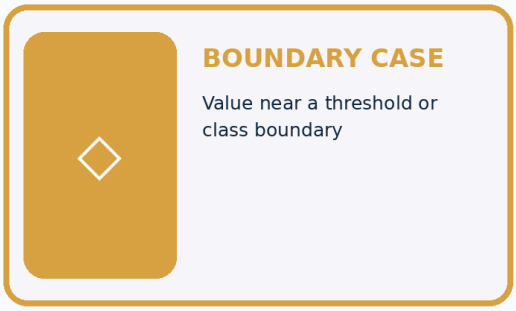
> Compare the supplied cases for Train, Validation, Test. Identify the detail that changes the conclusion.

EVALUATION DESIGN

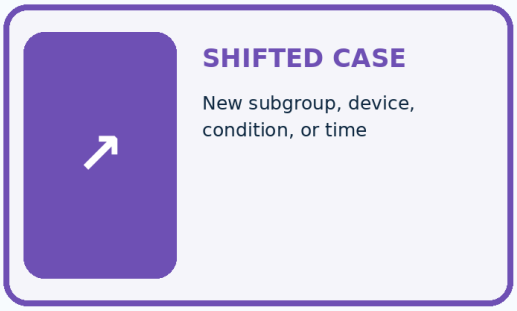
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule.

Predicted result

Prediction: Select A using the predeclared simpler-model tie-break.

Why the result follows

Reason: The **validation** errors tie, so the stated second a = ry rule decides without consulting final data.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Train, Validation, Test to explain a concrete decision, rather than only memorising definitions.

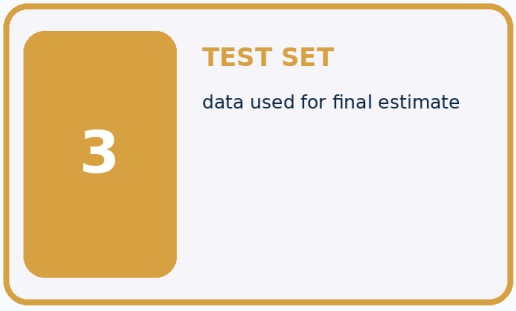
VOCABULARY

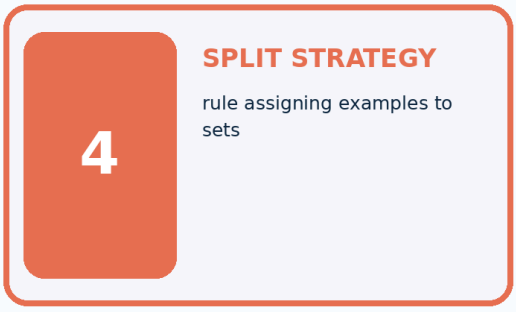
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
training set	data used to fit parameters	
validation set	data used to choose settings	
test set	data used for final estimate	
split strategy	rule assigning examples to sets	

MISCONCEPTION TO REPAIR Three random folders are enough even when photos of one plant appear in each.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Validation	Development evidence used to choose between options
Hyperparameter	A model setting chosen outside its fitted parameters
Selection rule	A declared method for choosing an option
Test set	Held-out evidence reserved for a final evaluation

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Train, Validation, Test, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 24 • WORKED EVIDENCE

1. Fit two candidate trees on training rows.
2. Choose depth and threshold using validation metrics and error slices.
3. Freeze preprocessing, features, model, and decision threshold.
4. Evaluate once on test; do not return to tune and still call it final.

Step 1

STEP 1 Fit two candidate trees on **training** rows.

Step 2

STEP 2 Choose depth and threshold using **validation** metrics and error slices.

Step 3

STEP 3 Freeze pre**process**ing, features, model, and decision threshold.

Step 4

STEP 4 Evaluate once on test; do not return to tune and still call it final.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Validation and model selection at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Training	Fit each candidate	No test labels
Validation	Choose A by rule	Record every attempt
Final test	Frozen model only	Report after selection

A tie policy prevents a convenient choice after inspecting results. Equal validation scores do not prove equal future performance. Record candidate settings, split identifiers, preprocessing and the full comparison, including rejected options.

Where should parameter fitting, setting selection and final reporting occur? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Train, Validation, Test

Assign a separate job to **training**, **validation** and final test **evidence**.

The facts we will use

> **Supplied facts**
>
> Candidates A and B have **training** MAE1 and 0 cm; **validation** MAE2 and 3 cm. **Selection rule** is smallest **validation** MAE.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/24

Train, Validation, Test

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit on training

Both candidates are built using training data; B fits it more closely.



2. Select on validation

Choose A because2 cm is below3 cm.



3. Reserve final test

Freeze A before scoring final cases; do not select again using their labels.

CHANGE ONE THING

A and B both have validation MAE2 cm; A is simpler under a predeclared tie rule.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit on training	Both candidates are built using training data; B fits it more closely.
2. Select on validation	Choose A because2 cm is below3 cm.
3. Reserve final test	Freeze A before scoring final cases; do not select again using their labels.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

Keep the stated limit

Read the labelled picture: Why does B's zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Train, Validation, Test claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Three random folders are enough even when photos of one plant appear in each. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Three random folders are enough even when photos of one plant appear in each.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A low score is meaningful only with its unit, metric, evaluated population and baseline. Decide how to handle ties and missing **prediction**s before comparison.

If no appropriate final test exists, state that limitation and keep claims about **validation**.

Why is the winning validation score not a guaranteed future score?

STUDENT LESSON · 8 OF 18

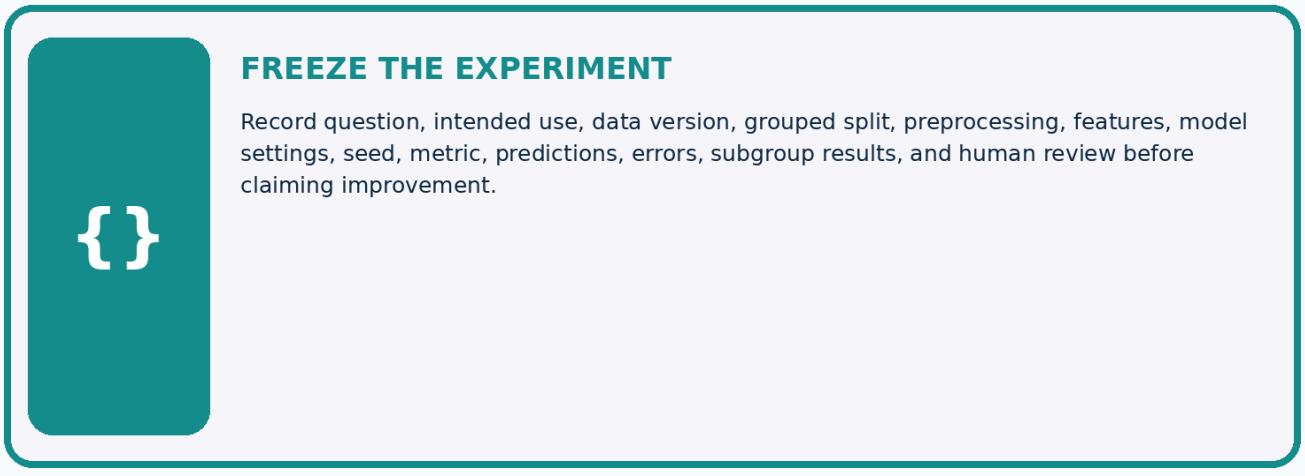
Practise together

Practise Train, Validation, Test with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

If you inspect **validation** errors and invent candidate D, that is another development attempt. Log it. Repeated experiments can adapt to peculiarities of **validation**. Do not rename **validation** as an untouched test.

More rows alone do not repair duplicated groups or future information leaking into **input**s.

Practise with less help: Assign data roles

Use workbook W2 for the connected example "**Validation** and model selection". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Validation** and model selection. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why does B’s zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Train, Validation, Test. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Role Assignment: sort twenty actions into train/validation/test, then group related examples before splitting.

1. Write the hypothesis and expected **evidence** before running.

2. Role Assignment: sort twenty actions into train/**validation**/test, then group related examples before splitting.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

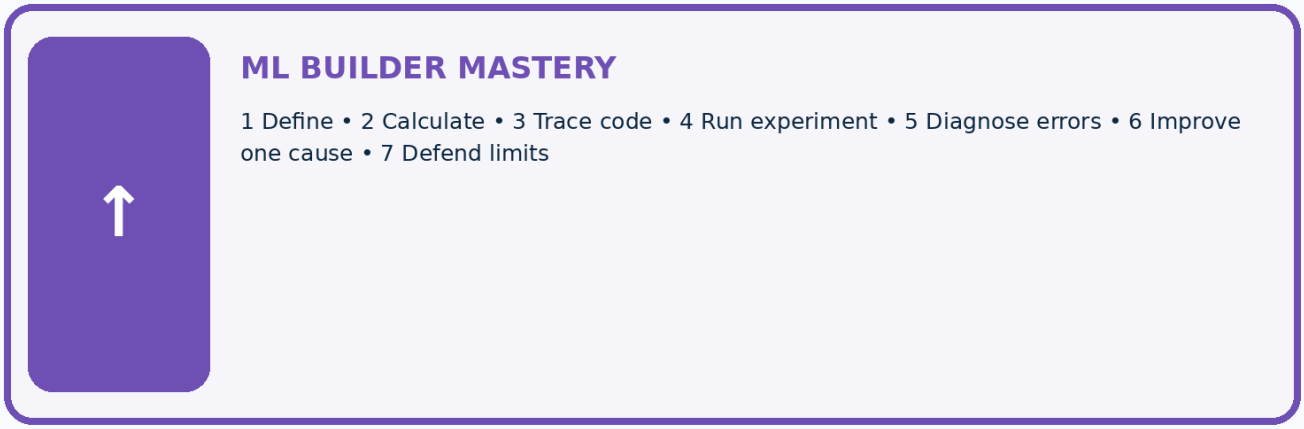
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Train, Validation, Test. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **training set**, **validation set**, **test set**, **split strategy**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

> TRACE Explain the data, code or decision path.

> DIAGNOSE Three random folders are enough even when photos of one plant appear in each.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

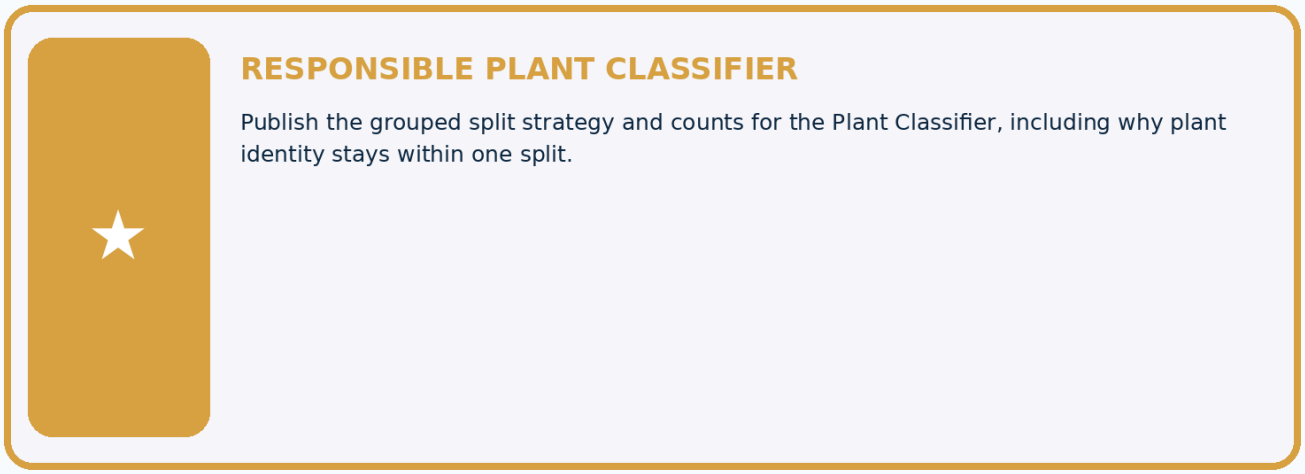
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Publish the grouped **split strategy** and counts for the Plant Classifier, including why plant identity stays within one split.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

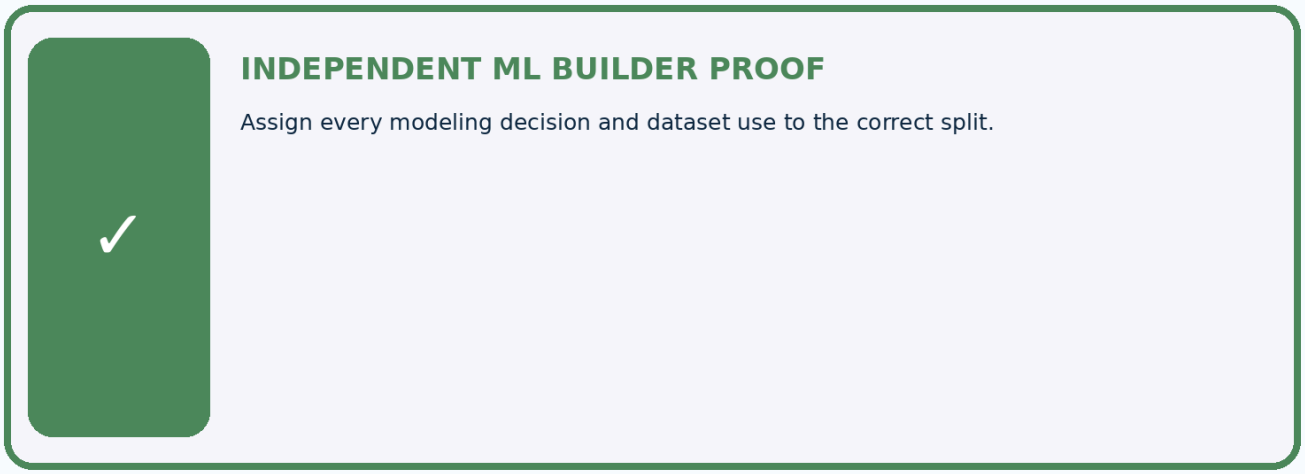
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Assign every modeling decision and dataset use to the correct split.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Metrics Match Consequences.

Study a complete case

> Revisit the complete Train, Validation, Test case for supported practice. Repeated examples do not become fresh assessments.

Week 24 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Split 100 independent leaves into 60 train,20 **validation**,20 test.

The question

Explain roles and pre**process**ing.

Worked explanation

Train fits parameters and learned transforms; **validation** chooses settings; final test evaluates the selected system. Group related records first.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Three partitions have three different jobs

Evidence or stage	Value or result
Train, fit parameters	60
Validation, choose settings	20
Test, evaluate selected system	20

Split 100 independent leaves into 60 train,20 **validation**,20 test.

Explain roles and pre**process**ing.

Train fits parameters and learned transforms; **validation** chooses settings; final test evaluates the selected system. Group related records first.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Training errors P 2 Q 0 R 1; **validation** errors P 4 Q 6 R 4. Rule: smallest **validation** MAE, then simpler; P is simpler than R. Choose and state the test action.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 24 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Assign roles: set A fits model parameters; B selects between two models; C is opened once after choices are fixed. Which is train, **validation** and test?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Publish the grouped **split strategy** and counts for the Plant Classifier, including why plant identity stays within one split.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

You have 30 leaves with 3 views each. Allocate 18 leaves to train,6 to **validation**,6 to test. How many images are in each? Where is a learned normalisation mean fitted?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Train, Validation, Test, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Assign a separate job to **training**, **validation** and final test **evidence**.

By the end, I can	How I will show it
Select a model using a declared validation rule	Explain the deciding step, show a trace or test, and state one limit.
Keep training, validation and test roles separate	Explain the deciding step, show a trace or test, and state one limit.
Recognize repeated validation as development	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Train, Validation, Test without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Select a model using a declared **validation** rule

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Assign a separate job to **training**, **validation** and final test **evidence**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Freeze A before scoring final cases; do not select again using their labels.

1. Reserve final test

2. Fit on **training**

3. Select on **validation**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reserve final test Freeze A before scoring final cases; do not select again using their labels. Read the picture in order.

Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Select A using the predeclared simpler-model tie-break.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Select A using the predeclared simpler-model tie-break. The **validation** errors tie, so the stated second a = ry rule decides without consulting final data.

Repair a missing building block

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised version?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as4.67 cm misleading?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

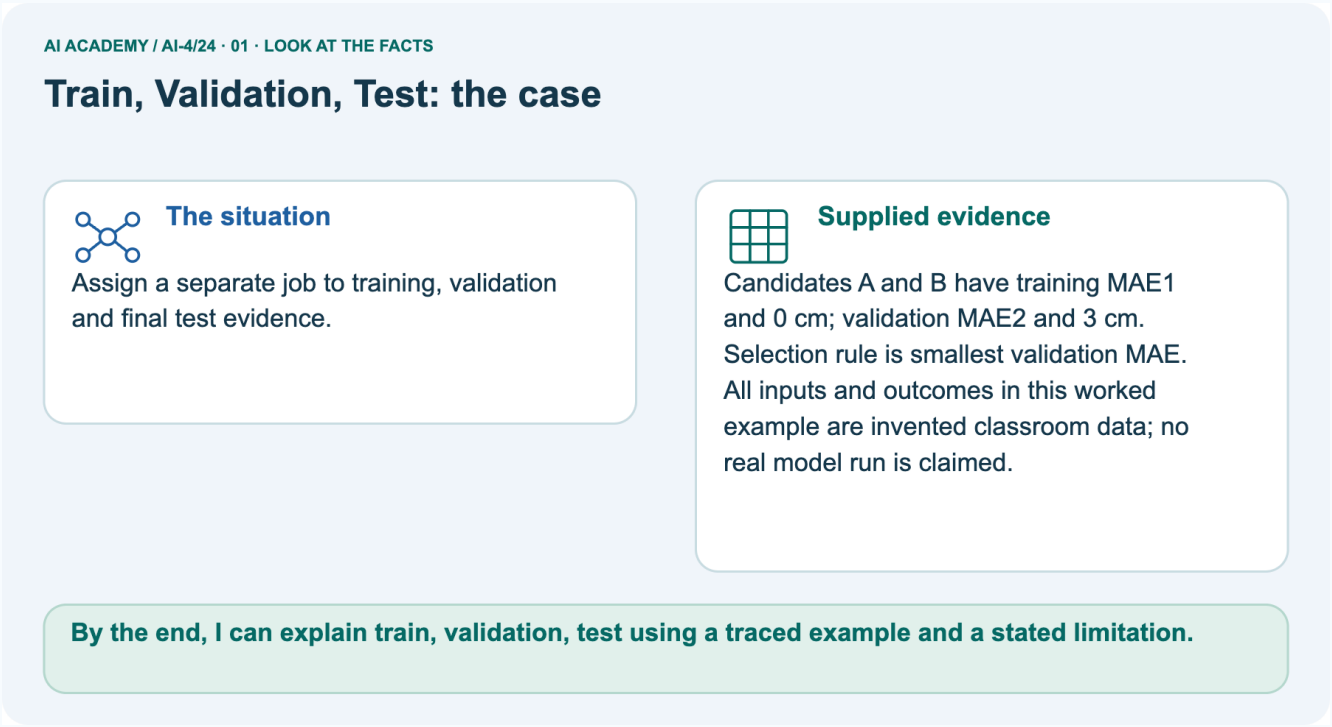
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Train, Validation, Test graphic using its labels, deciding evidence and relevant limitation.

01. Train, Validation, Test: the case



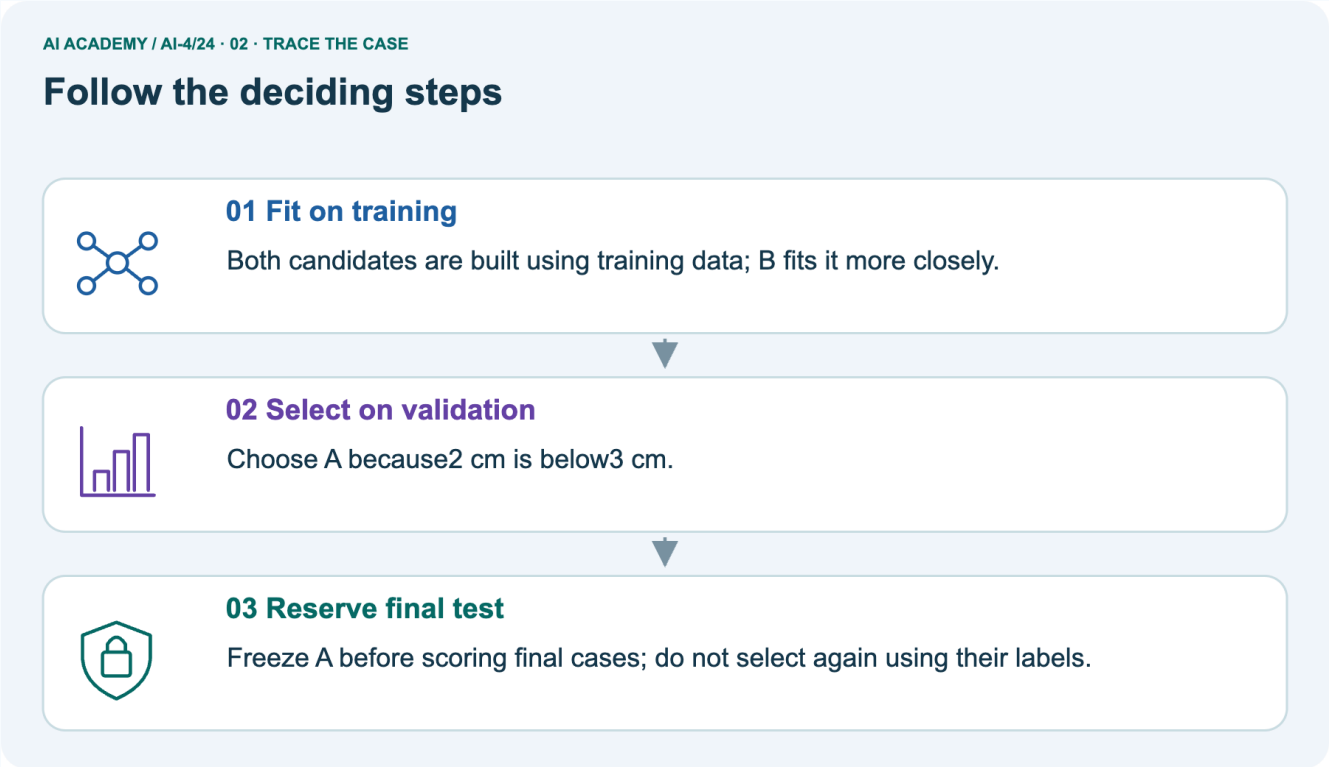
Follow the labelled picture with the matching explanation. Open the vector diagram

Assign a separate job to **training**, **validation** and final test **evidence**. Supplied facts: Candidates A and B have **training** MAE1 and 0 cm; **validation** MAE2 and 3 cm. **Selection rule** is smallest **validation** MAE.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit on **training**: Both candidates are built using **training** data; B fits it more closely. → Select on **validation**: Choose A because2 cm is below3 cm. → Reserve final test: Freeze A before scoring final cases; do not select again using their labels.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Validation and model selection at a glance

AI ACADEMY / AI-4/24 · 03 · INSPECT THE LESSON MODEL

Validation and model selection at a glance

Part	Evidence or action	Interpretation
Training	Fit each candidate	No test labels
Validation	Choose A by rule	Record every attempt
Final test	Frozen model only	Report after selection

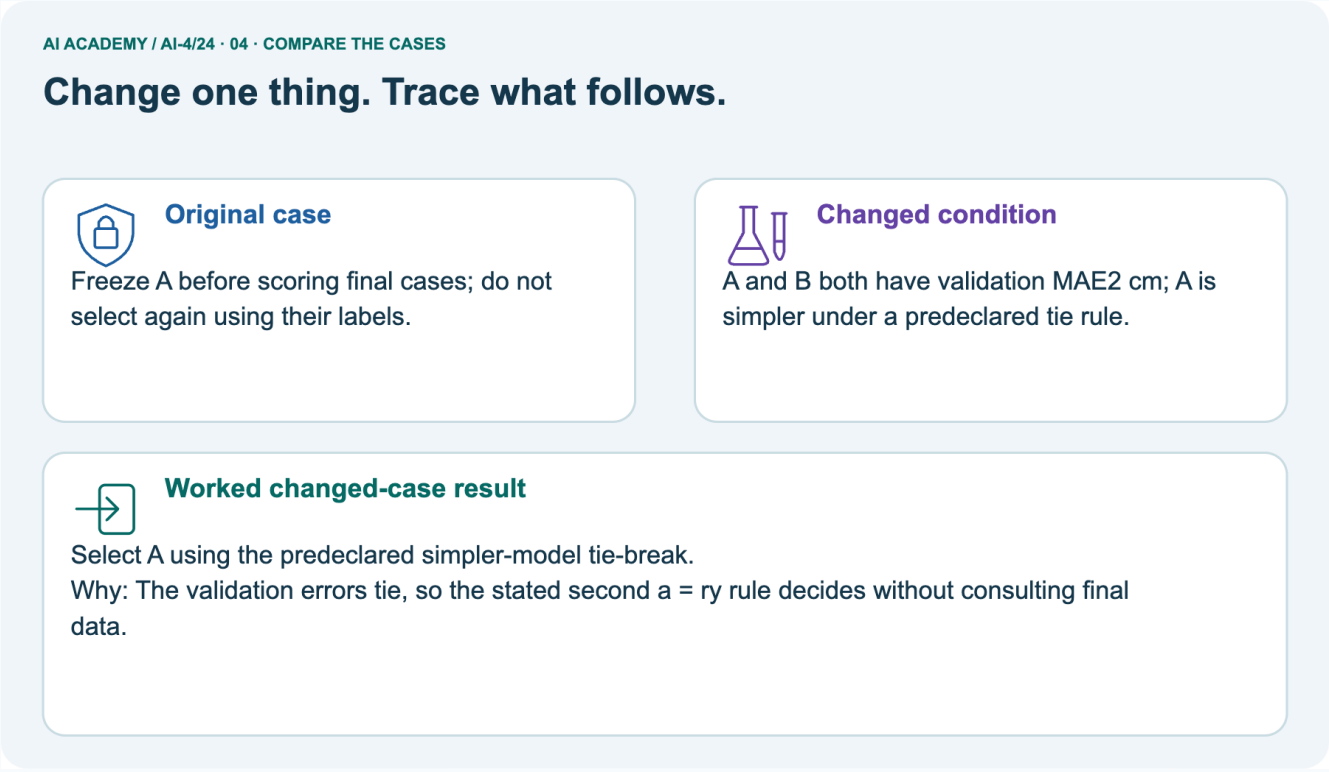
A tie policy prevents a convenient choice after inspecting results. Equal validation scores do not prove equal future performance. Record candidate settings, split identifiers, preprocessing and the full comparison, including rejected options.

Follow the labelled picture with the matching explanation. Open the vector diagram

Validation and model selection at a glance. **Training** | Fit each candidate | No test labels. **Validation** | Choose A by rule | Record every attempt. Final test | Frozen model only | Report after selection

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



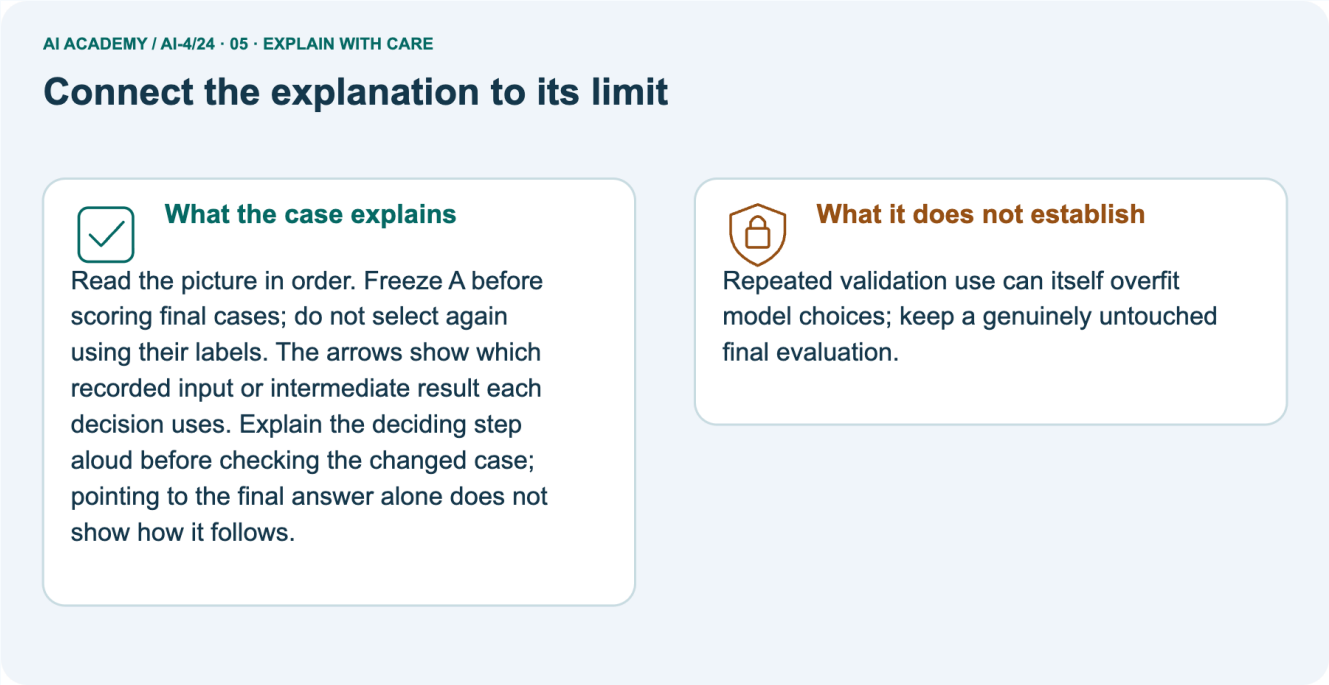
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Freeze A before scoring final cases; do not select again using their labels. Changed condition: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule.

Expected result: Select A using the predeclared simpler-model tie-break. Reason: The **validation** errors tie, so the stated second a = ry rule decides without consulting final data.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

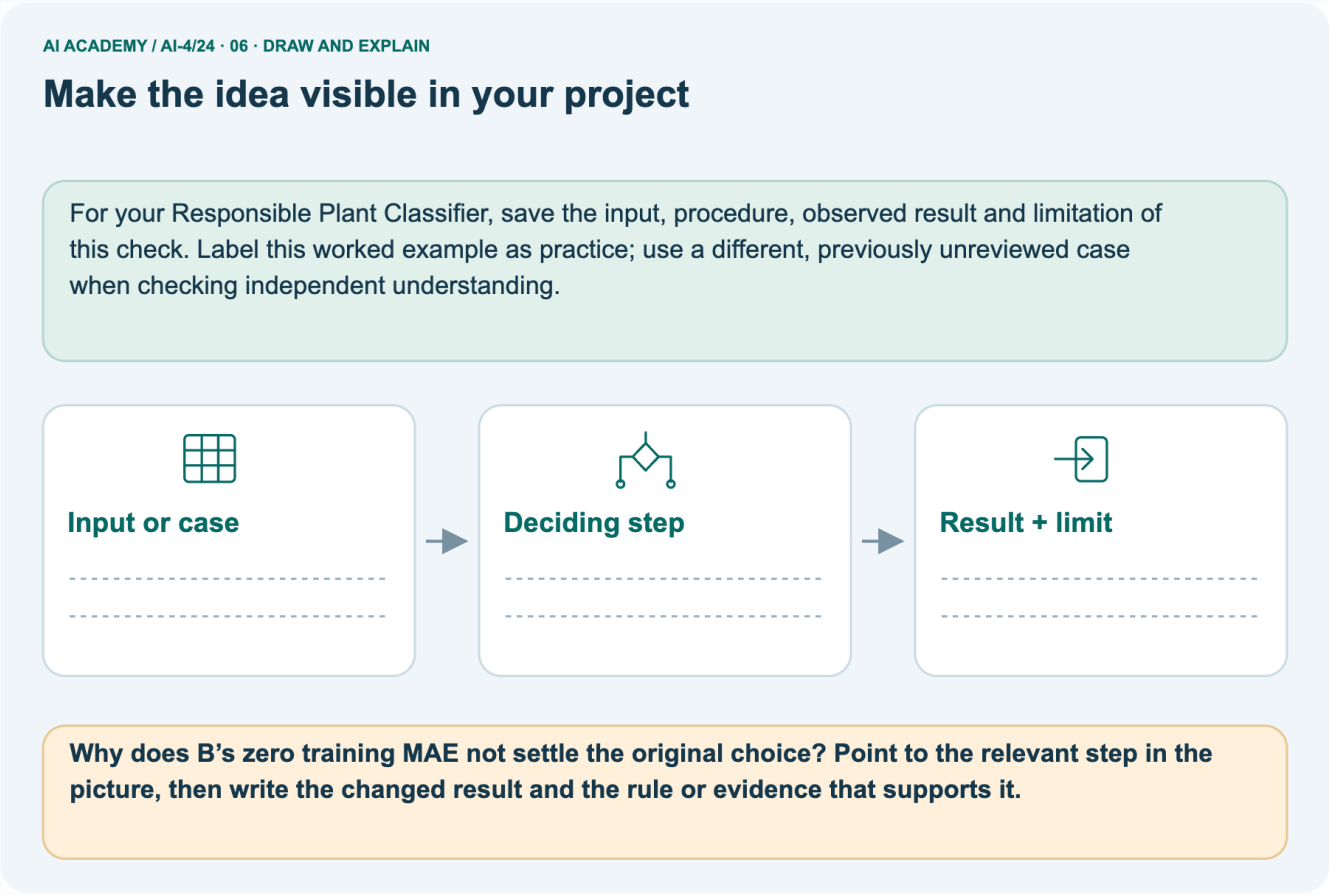
Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why does B's zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.